

1. Introduction

Digital forensics is an essential part of modern cybersecurity, enabling investigators to identify, preserve, and analyze digital evidence. This process allows investigators to reconstruct system activity, determine what actions occurred, and support investigations involving security breaches or unauthorized access (National Institute of Standards and Technology [NIST], 2006). As organizations and individuals increasingly rely on digital systems for storing and managing sensitive information, the ability to analyze data at rest has become increasingly important. Data at rest refers to information stored on physical devices such as hard drives or solid-state drives, where traces of user activity can remain even after files are deleted or altered (NIST, 2006).

The New Technology File System (NTFS), developed by Microsoft and introduced in 1993 with Windows NT 3.1, is the primary file system used in modern Windows operating systems (Microsoft, 2023). In addition to organizing and storing files, NTFS maintains structured metadata that records detailed information about file system activity, including timestamps, file attributes, and transaction records (Microsoft, 2023). This metadata can serve as valuable forensic evidence as it provides insight into how a system has been used and what actions have taken place, particularly in cases involving unauthorized access or suspicious activity.

Among the most significant NTFS artifacts are the Master File Table (MFT) and the \$LogFile, which provide related but distinct perspectives on file system activity. The MFT acts as a centralized database that stores metadata for every file and directory on a drive's volume, while the \$LogFile functions as a journaling mechanism that records changes to file system metadata, creating a record of file system transactions over time (Microsoft, 2023). When examined individually, these artifacts provide useful information, but when analyzed together, they enable investigators to reconstruct detailed timelines of file activity and identify inconsistencies that may indicate tampering or anti-forensic techniques.

2. Background

NTFS organizes and manages data through structured metadata and internal records. Rather than simply storing files as raw data, NTFS maintains detailed information about them in a standardized format, allowing the operating system (OS) to efficiently track and manage file system activity (Microsoft, 2023). This design is particularly important in digital forensics, as it ensures that system activity is consistently recorded within the file system itself, rather than relying solely on higher-level logs, which are more likely to be altered or removed. Because NTFS embeds metadata directly within its core structures, investigators can often recover evidence even when user-level logs have been cleared or manipulated.

At the core of NTFS is the concept of attributes, which define all aspects of a file including name, size, location, permissions, and timestamps. These attributes are stored within the MFT, with each file or directory being associated with a single MFT record (Microsoft, 2023). MFT records contain multiple attributes, including the \$STANDARD_INFORMATION and \$FILE_NAME attributes, both of which store timestamp data. These timestamps, commonly referred to as MAC times, record when a file was created, modified, accessed, or when its metadata was changed. Because these timestamps are maintained separately across different attributes, inconsistencies between them can provide important forensic indicators. This is because \$STANDARD_INFORMATION timestamps reflect system-level changes, while \$FILE_NAME timestamps reflect directory-level changes, resulting in the two sets often showing contrasting times even for the same file due to operations such as moving, renaming, or metadata updates.

In addition to metadata storage, NTFS implements a journaling system through the \$LogFile, which records file system transactions by logging intended changes to metadata before they are fully written to the disk. This includes both redo and undo operations, allowing the system to complete or reverse the changes in the event of a failure (Microsoft, 2023). From a forensic perspective, this behavior creates a partial history of file system activity, which captures operations such as file creation, deletion, and

modification even when those changes are no longer visible in the current file system state. However, because the \$LogFile is a circular log with a limited size, older entries are overwritten, meaning it only preserves a partial, time-sensitive record of activity.

Another key component of NTFS is the Update Sequence Number Journal (\$UsnJrnl), which tracks file-level changes over time. While the \$LogFile focuses on maintaining file system consistency, the \$UsnJrnl records ongoing changes such as file modifications, renaming and deletion events (Microsoft, 2023). Unlike the \$LogFile's low-level transactional focus, the \$UsnJrnl provides a higher-level and longer-term record of the file system changes, making it particularly useful for validating or supplementing evidence from other NTFS artifacts. This artifact provides an additional source of information that investigators use to validate findings from the MFT and \$LogFile. Together, these NTFS components form the foundation for how artifact correlation supports forensic analysis.

3. Tool or Technique Description

The primary technique used in NTFS forensic analysis is the correlation of artifacts, specifically between the MFT, \$LogFile, and other supporting data such as the \$UsnJrnl. Rather than relying on a single artifact, investigators compile multiple sources from the file system to cross-reference data across related artifacts.

At a high level, this technique involves comparing the static metadata that is stored in the MFT with transactional records preserved by the \$LogFile. As previously described, the MFT provides information about the current or most recent state of files and directories, including attributes such as file names, timestamps, and storage details. While the MFT reflects the current state of the file system, the \$LogFile records changes to file system metadata as they occur, capturing operations such as file creation, modification, renaming, and deletion (Microsoft, 2023).

The correlation of NTFS artifacts involves examining timestamp data, log records, and attribute changes across multiple sources. Investigators compare MAC times in the MFT with entries in the \$LogFile to determine whether recorded timestamps accurately reflect file system activity. If inconsistencies are identified, this may indicate that timestamps have been altered, such as through timestamping (SANS Institute, 2015). Additionally, the \$LogFile may contain records of file operations that are no longer present in the MFT, providing evidence of deleted or modified files (Microsoft, 2023).

In addition to the MFT and \$LogFile, the \$UsnJrnl can be used to supplement this analysis by providing a continuous record of file-level changes. This allows investigators to validate findings across multiple sources and identify activity that may not have been fully captured by a single artifact. By correlating these artifacts, investigators can construct a more reliable timeline of system activity, reducing the likelihood of misinterpretation caused by incomplete or overwritten data.

4. Forensic Application

In practical forensic investigations, NTFS artifact correlation is used to answer specific questions about user behavior and system activity, particularly when evidence has been altered or partially removed by an attacker or due to the time-sensitive nature of the \$LogFile. Investigators often rely on this technique when analyzing suspicious file access, determining whether files were intentionally deleted, or verifying the authenticity of file timestamps. For example, if a system is suspected of being used to stage or remove sensitive data, investigators can analyze NTFS artifacts to determine when files were created, accessed, or removed, even if the files are no longer present. By examining the residual metadata of file system records, they can identify patterns of rapid or repeated file access followed by deletion, which may indicate data exfiltration or attempts to conceal unauthorized activity (SANS Institute, 2015).

This method is also applied in situations where user actions need to be verified or disputed. In cases involving insider threats or unauthorized access, investigators may use NTFS artifacts to confirm

whether a user interacted with specific files at a given time. Even if logs at the OS or application level have been cleared, the underlying file system artifacts can contain trace evidence of those interactions (Microsoft, 2023). Additionally, when inconsistencies are found between artifacts, such as mismatched timestamps or missing records, these can indicate that attempts were made to alter or manipulate evidence (SANS Institute, 2015).

5. Strengths and Limitations

One of the primary strengths of NTFS artifact correlation is its ability to provide a reliable and multi-layered view of file system activity. By analyzing multiple artifacts such as the MFT, \$LogFile, and \$UsnJrnl together, investigators are able to cross-reference data and validate findings across multiple sources. This reduces the likelihood of misinterpretation and makes it more difficult for malicious activity to go unnoticed. Additionally, because NTFS records metadata directly within its internal structures, relevant evidence can often persist even after the files have been deleted or higher-level logs have been cleared (Microsoft, 2023). This makes NTFS artifact analysis particularly valuable in cases where attackers attempt to conceal their actions.

However, there are several limitations and challenges when analyzing NTFS. One significant limitation is that the \$LogFile operates as a circular log with a finite size, meaning that older entries are eventually overwritten and are no longer available for analysis. Similarly, the \$UsnJrnl can also be truncated or disabled, reducing the amount of historical data available to investigators. Another challenge involves timestamp inconsistencies, which may occur naturally due to system behavior, caching, or clock-synchronization issues, making it difficult to distinguish between real system behavior and intentional manipulation. Advanced anti-forensic techniques, such as timestomping, can alter file metadata to obscure activity, requiring investigators to rely on multiple artifacts to distinguish inconsistencies (SANS Institute, 2015). As a result, while NTFS artifact correlation is a powerful method, it must be applied carefully and in conjunction with other forensic techniques to ensure accurate conclusions.

6. Conclusion

NTFS artifact analysis plays a critical role within modern digital forensics by providing investigators with the ability to examine and interpret file system activity at a detailed level. By correlating multiple artifacts such as the MFT, \$LogFile, and \$UsnJrnl, investigators can identify inconsistencies, verify user actions, and uncover evidence that may not be visible through higher-level system logs. This approach allows for a more accurate reconstruction of events, especially in cases where data has been altered, deleted, or intentionally concealed.

While this technique presents certain limitations, such as incomplete data due to log overwriting and the possibility of manipulated timestamps, it remains a highly effective method when applied correctly. The ability to cross-reference multiple sources of evidence strengthens the reliability of forensic conclusions regarding suspicious or malicious activity, as well as normal system processes. As digital systems continue to evolve, understanding and applying NTFS artifact correlation will remain an essential skill for forensic professionals.

7. References

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